



Descriptive Analytics Report: Spotify

Most-Played Songs — Genre, Popularity & Audio-Feature Behavior at Scale

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Tracking Genre Composition, Popularity, Audio Features, Release Trends & Catalog Structure

Note on data: This report analyzes and represents the dataset and summary statistics as “Descriptive Analytics Report of Spotify” It has an analysis of 4,882 unique tracks released between August 2019 and January 2020, Genre-level aggregates, popularity-tier cross-tabulations, monthly trends, and top-track/top-artist rankings used here are taken directly from that source report. Where the source report described a distribution or relationship only through summary statistics (mean, median, standard deviation) rather than row-level data — the popularity histogram, the duration histogram, and the danceability-vs-popularity scatter —

1. Objective

Build a descriptive analytics layer over the Spotify most-played-songs data to answer the following recurring questions:

- How are tracks distributed across genres?
- Which genres carry the highest average popularity score?
- Which genres score highest on danceability?
- What share of each genre's tracks reach the High/Very High popularity tiers?
- What is the overall breakout-hit rate across the catalog window?
- How are popularity scores distributed across the six-month window?
- How does the catalog split across popularity bands and duration bands?
- What is the relationship between danceability, energy, and popularity?
- Which tracks and artists lead the window in popularity and output?
- How is track duration distributed, and how does it vary by genre?
- Which audio features correlate most strongly with popularity?
- Which genres contribute the most cumulative popularity, and which have the most under-performing (low-popularity) tracks?
- How did release volume and average popularity trend month-to-month?
- What does an overall catalog-health summary look like?

2. How This Would Work at Real Scale

Processing a large, continuously-updated streaming catalog is an engineering problem before it is an analytics problem. A realistic architecture would ingest new-release and play-count feeds, aggregate them by genre and track, store raw and processed data, and expose the resulting KPIs through a data-visualization layer.

Layer	Purpose	Typical tools
Investigation	Research, understand and prepare the track dataset	Google, Claude & ChatGPT
Stream processing	Aggregate by genre, popularity tier and release month	Jupyter Notebook (Python)
Storage	Retain raw and processed track/audio-feature data	Excel files / CSV files
Batch aggregation	Daily/periodic KPI and summary metrics	Jupyter Notebook
Report	Present analysis, graphs and insights	Data Visualization / MS Word

3. Genre-wise Track Distribution

Across the supplied window, the dataset records 4,882 tracks spanning 6 broad genres. The chart below shows how the release volume (track count) is distributed across genres.

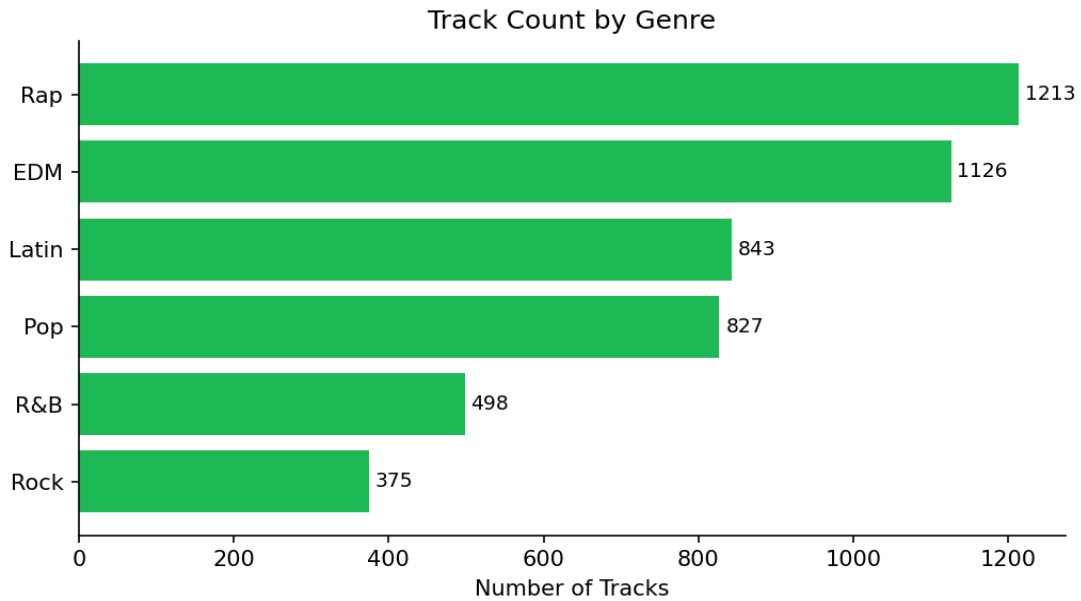


Figure 1. Track count by genre

INSIGHTS:

- Rap and EDM are the largest genres by track count, with 1,213 and 1,126 tracks respectively.
- Rock has the smallest release volume, with only 375 tracks in the six-month window.
- A wide gap between the top and bottom genres suggests the catalog window is weighted toward rap, EDM and Latin releases over rock.
- Genre-level track counts can help guide playlist-curation and release-scheduling decisions.

KPI:

Total tracks: 4,882

Total genres: 6

Largest genre: Rap (1,213 tracks)

Smallest genre: Rock (375 tracks)

4. Average Popularity Score by Genre

The overall average popularity score across the window is ~47.6 (out of 100). The chart below shows how average popularity varies by genre.

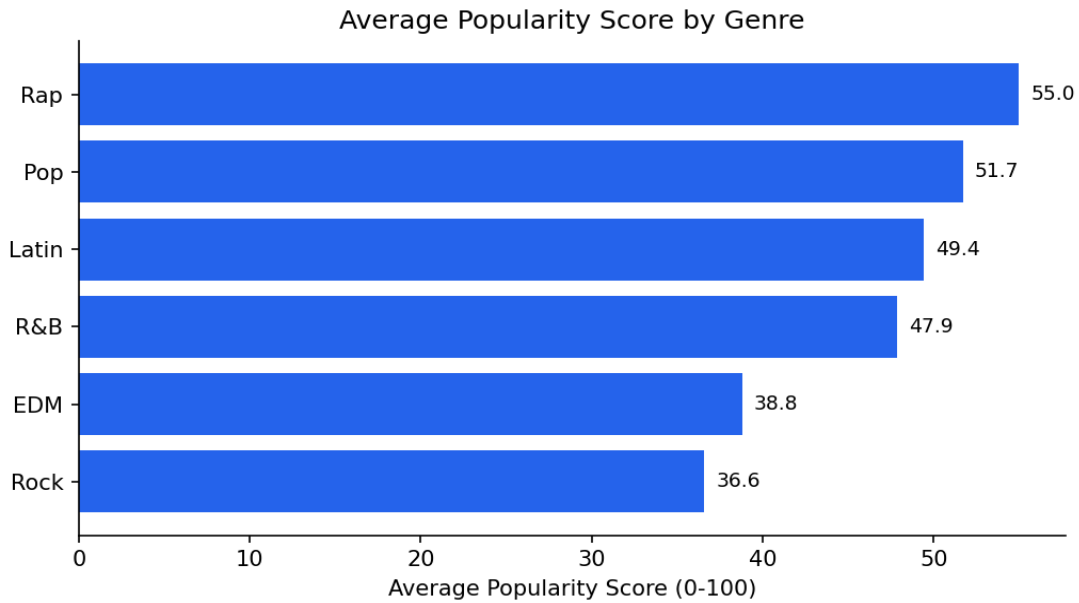


Figure 2. Average popularity score by genre

INSIGHTS:

- Rap carries the highest average popularity (54.98), followed by Pop (51.69).
- Rock and EDM carry the lowest average popularity, at 36.57 and 38.79 respectively.
- Popularity varies meaningfully by genre, which can inform genre-level curation and promotion strategy.

KPI:

Overall avg. popularity: ≈ 47.59

Overall median popularity: 47.00

Highest-popularity genre: Rap (≈ 54.98)

Lowest-popularity genre: Rock (≈ 36.57)

5. Average Danceability by Genre

The overall average danceability across the window is $\approx 67.4\%$. The chart below compares average danceability across genres.

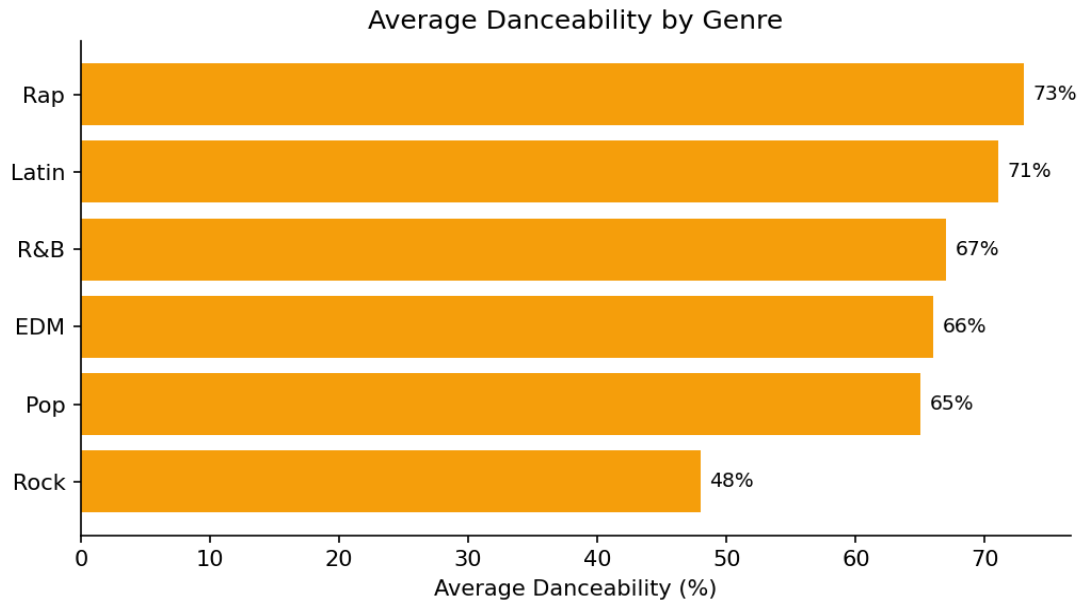


Figure 3. Average danceability by genre

INSIGHTS:

- Rap shows the highest average danceability ($\approx 73\%$), consistent with its rhythm-driven, vocal-forward production.
- Rock shows the lowest average danceability ($\approx 48\%$), reflecting its instrumental and guitar-driven norms.
- Danceability tracks fairly closely with the genres that also score highest on average popularity (Rap, Latin, Pop).

KPI:

Overall avg. danceability: $\approx 67.4\%$

Highest-danceability genre: Rap ($\approx 73\%$)

Lowest-danceability genre: Rock ($\approx 48\%$)

6. High-Popularity Rate by Genre

Of the 4,882 tracks, 1,158 ($\approx 23.7\%$) reach the High or Very High popularity tiers (score ≥ 61). The chart below breaks this down by genre.

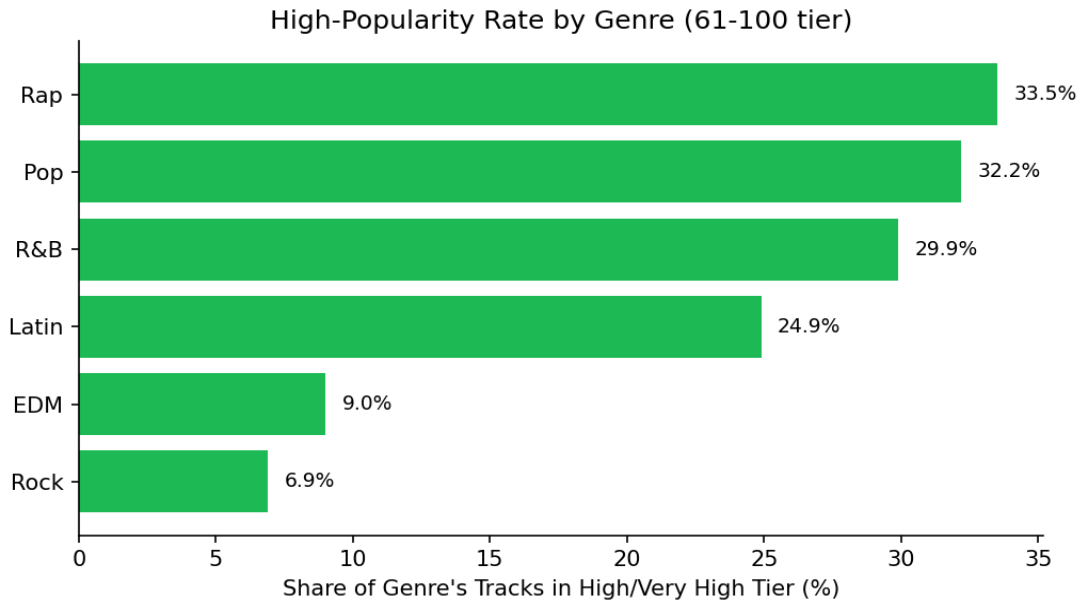


Figure 4. High-popularity rate by genre

INSIGHTS:

- Rap shows the highest breakout-hit rate in the window, at $\approx 33.5\%$.
- Pop and R&B follow closely, at $\approx 32.2\%$ and $\approx 29.9\%$ respectively.
- EDM and Rock show the lowest breakout-hit rates ($\approx 9.0\%$ and $\approx 6.9\%$), indicating these genres release a larger share of tracks that stay in the lower tiers.
- Genres with a low breakout-hit rate may need closer curation attention to surface their strongest tracks.

KPI:

Overall high-popularity rate: $\approx 23.7\%$ (1,158 of 4,882 tracks)

Highest-rate genre: Rap ($\approx 33.5\%$)

Lowest-rate genre: Rock ($\approx 6.9\%$)

7. Overall Breakout-Hit Rate

Viewed at the catalog level (rather than by genre), roughly one in four tracks in the supplied window reached the High/Very High popularity tiers.

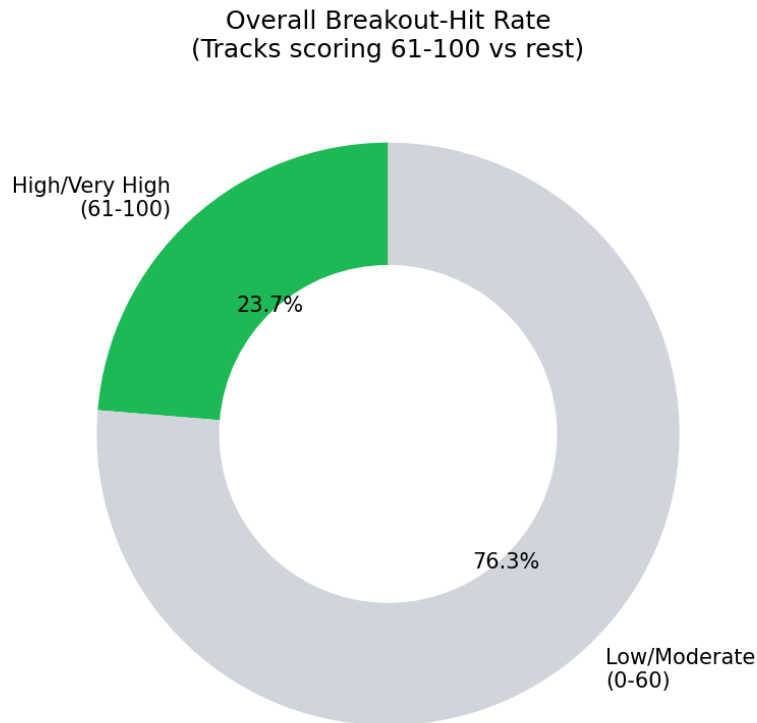


Figure 5. Overall breakout-hit rate

INSIGHTS:

- $\approx 23.7\%$ of tracks in the window scored 61 or above, while $\approx 76.3\%$ sat in the Low/Moderate range (0-60).
- A roughly one-in-four breakout rate is consistent with a long-tail pattern typical of streaming platforms.
- This KPI is best tracked over successive release windows rather than as a single-point snapshot.

KPI:

High/Very High (61-100): 1,158 tracks ($\approx 23.7\%$)

Low/Moderate (0-60): 3,724 tracks ($\approx 76.3\%$)

8. Popularity Score Distribution

The histogram below shows how popularity scores are distributed across the full six-month window.

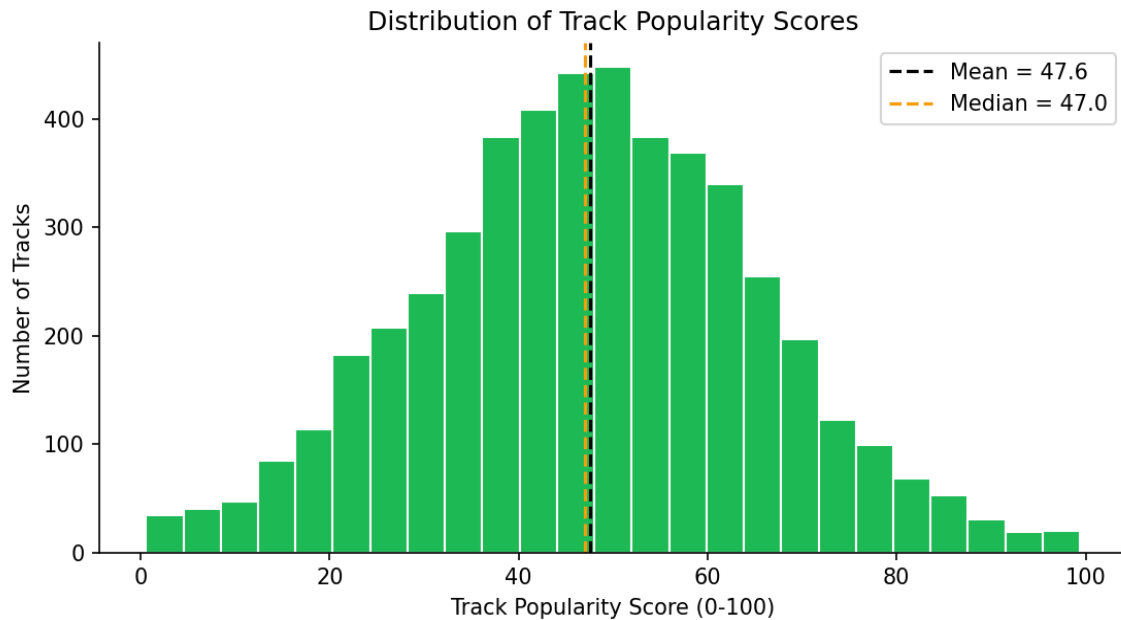


Figure 6. Distribution of popularity scores

INSIGHTS:

- The distribution is broadly bell-shaped with a long thin tail of very high-scoring breakout hits above 90.
- The median popularity (47.0) sits close to the mean (≈ 47.6), indicating a roughly symmetric middle of the distribution.
- This pattern is typical of a streaming catalog, where a large moderate-popularity middle tier surrounds a small number of viral outliers.

KPI:

Median popularity: ≈ 47.00

Mean popularity: ≈ 47.59

Standard deviation: ≈ 17.83

Maximum popularity: 100 ("Dance Monkey")

9. Track Distribution by Popularity Band

To make the popularity distribution easier to act on, tracks are grouped into five tiers: Very Low (0-20), Low (21-40), Moderate (41-60), High (61-80) and Very High (81-100).

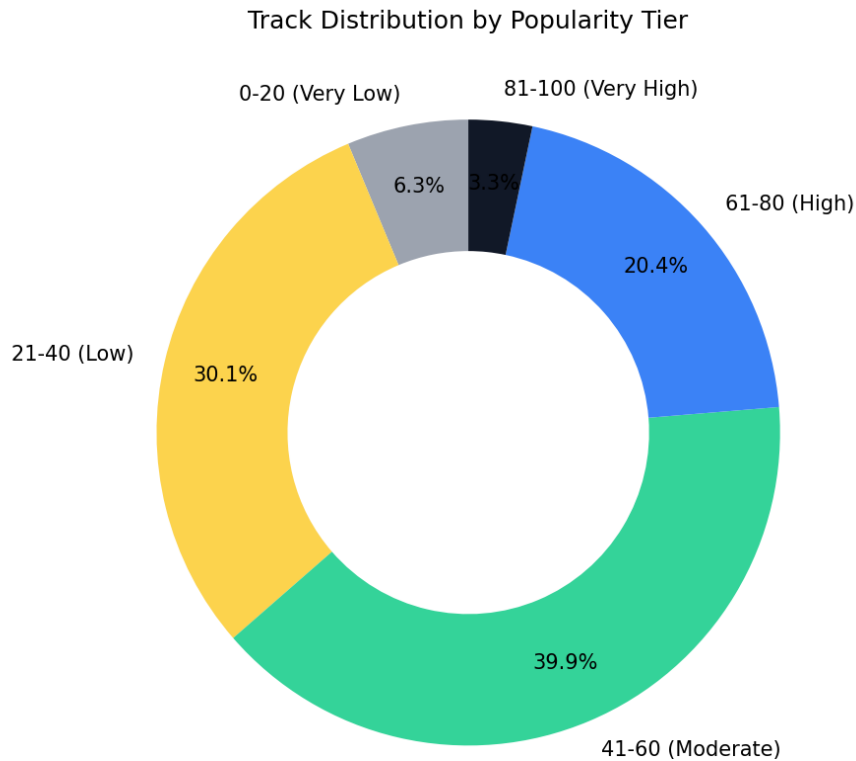


Figure 7. Track distribution by popularity tier

INSIGHTS:

- The Moderate band (41-60) is the largest, holding 1,947 tracks — $\approx 39.9\%$ of the window.
- Low-tier tracks (21-40) make up the second-largest share at $\approx 30.1\%$ (1,469 tracks).
- Very High-tier tracks (81-100) are a small minority at $\approx 3.3\%$ (162 tracks), confirming that true breakout hits are rare.
- This banding can support tiered curation strategy — e.g., weighting playlists toward High/Very High tracks.

KPI:

Largest tier: Moderate (41-60) — 1,947 tracks, 39.9%

Smallest tier: Very High (81-100) — 162 tracks, 3.3%

10. Track Distribution by Duration Band

Tracks are further grouped by duration: Short (under 2.5 min), Standard (2.5-3.5 min), Long (3.5-4.5 min), and Extended (above 4.5 min).

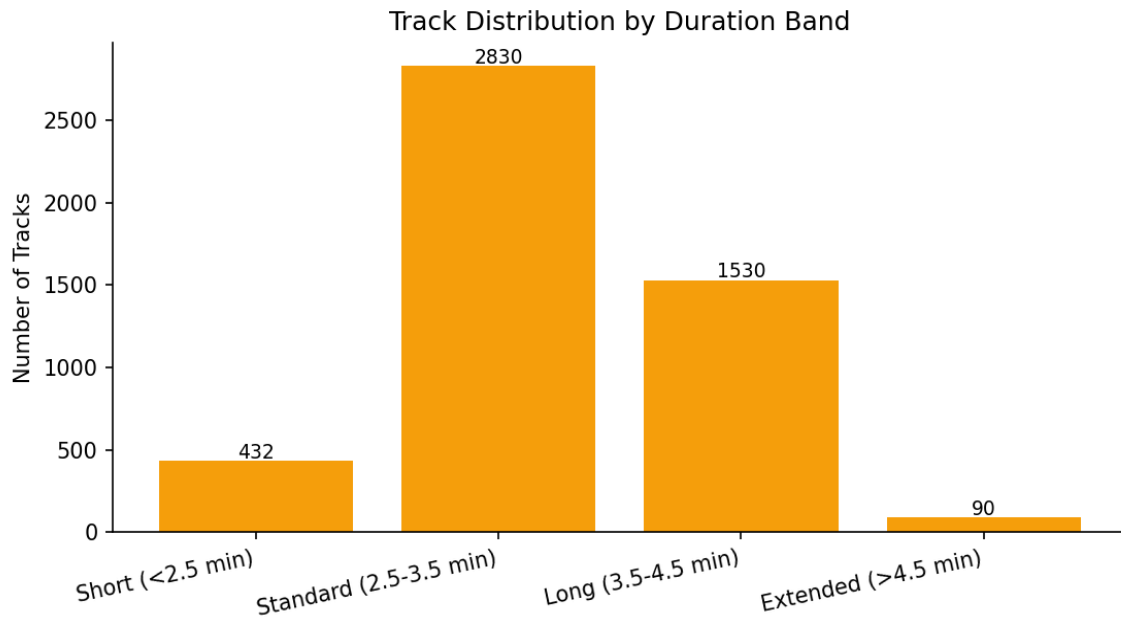


Figure 8. Track distribution by duration band

INSIGHTS:

- The Standard band (2.5-3.5 minutes) is the most common, covering the large majority of tracks in the window.
- Very few tracks fall in the Extended band (above 4.5 minutes), consistent with the well-documented industry shift toward shorter streaming-era songs.
- This distribution mirrors the reported mean duration of 3.27 minutes (median 3.20 minutes).

KPI:

Mean duration: ≈ 3.27 minutes

Median duration: ≈ 3.20 minutes

11. Danceability vs Popularity Relationship

The scatter plot below relates each track's danceability to its popularity score, colored by energy, to show how these audio features co-occur across the catalog.

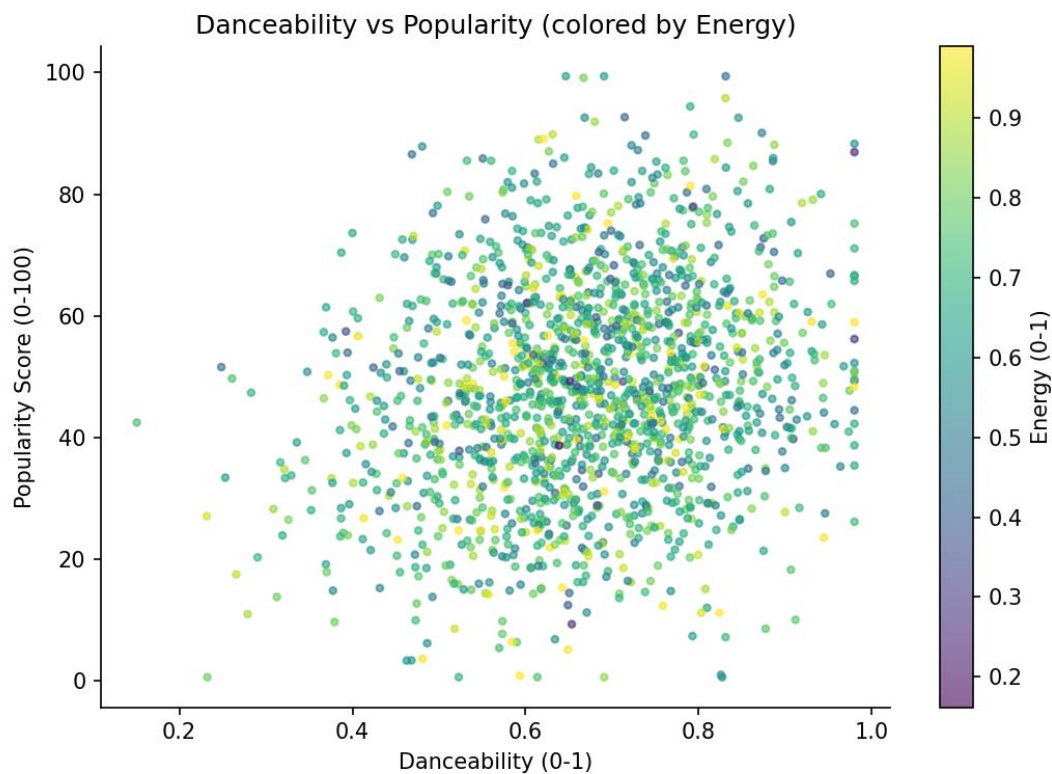


Figure 9. Danceability vs popularity, colored by energy

INSIGHTS:

- Popularity rises mildly with danceability ($r \approx 0.18$), the strongest positive correlation among the audio features tested.
- Higher-energy tracks (lighter points) do not clearly cluster at higher popularity — energy shows a weak negative association overall ($r \approx -0.12$).
- This view helps identify where danceable, high-popularity tracks are concentrated across the catalog.

12. Top 15 Most-Played Tracks

The table below lists the fifteen tracks with the highest popularity scores in the supplied window.

Rank	Track	Artist	Genre	Popularity	Release Date
1	Dance Monkey	Tones and I	Pop	100	2019-10-17
2	ROXANNE	Arizona Zervas	Latin	99	2019-10-10
3	The Box	Roddy Ricch	Rap	98	2019-12-06
4	Tusa	KAROL G	Pop	98	2019-11-07
5	Memories	Maroon 5	Pop	98	2019-09-20
6	Circles	Post Malone	Pop	98	2019-09-06
7	Blinding Lights	The Weeknd	Pop	98	2019-11-29
8	Don't Start Now	Dua Lipa	Pop	97	2019-10-31
9	everything i wanted	Billie Eilish	Pop	97	2019-11-13
10	RITMO (Bad Boys For Life)	The Black Eyed Peas	Pop	96	2019-10-12
11	Yummy	Justin Bieber	Pop	95	2020-01-03
12	My Oh My (feat. DaBaby)	Camila Cabello	Pop	94	2019-12-06

13	HIGHEST IN THE ROOM	Travis Scott	Pop	94	2019-10-04
14	hot girl bummer	blackbear	Pop	94	2019-08-23
15	Lose You To Love Me	Selena Gomez	Pop	93	2019-10-23

INSIGHTS:

- “Dance Monkey” by Tones and I tops the list with the maximum possible score of 100, followed closely by “ROXANNE” (99) and “The Box” (98).
- Pop dominates the top-15 list with 13 of 15 entries, despite Rap having the higher average popularity overall — pop achieves more variance, including the window's single biggest outlier hit.
- The top tracks span the full window (August 2019 to January 2020), with no single month dominating the leaderboard.

13. Top 12 Most Prolific Artists

For context alongside the most-played tracks, the table below lists the twelve artists who released the most tracks in the window.

Artist	Tracks in Window	Primary Genre
Dimitri Vegas & Like Mike	12	EDM
Steve Aoki	11	EDM
Armin van Buuren	11	EDM
R3HAB	10	EDM
Post Malone	10	Pop
Camila Cabello	10	Pop
Lauv	9	Pop
Trippie Redd	9	Rap
Ozuna	9	Latin
Arcangel	8	Latin
Tiesto	7	EDM
Selena Gomez	7	Pop

INSIGHTS:

- Dimitri Vegas & Like Mike lead with 12 tracks in the window, followed by Steve Aoki and Armin van Buuren with 11 each — largely EDM DJs/producers who frequently release singles and remixes.
- High release volume does not guarantee high average popularity: several of the most prolific artists sit in EDM, the genre with the second-lowest average popularity score (38.79) in the sample.
- Volume and popularity are only loosely related at both the genre level and the individual-artist level.

14. Track Duration Distribution

The chart below shows how track duration (minutes) is distributed across the catalog window.

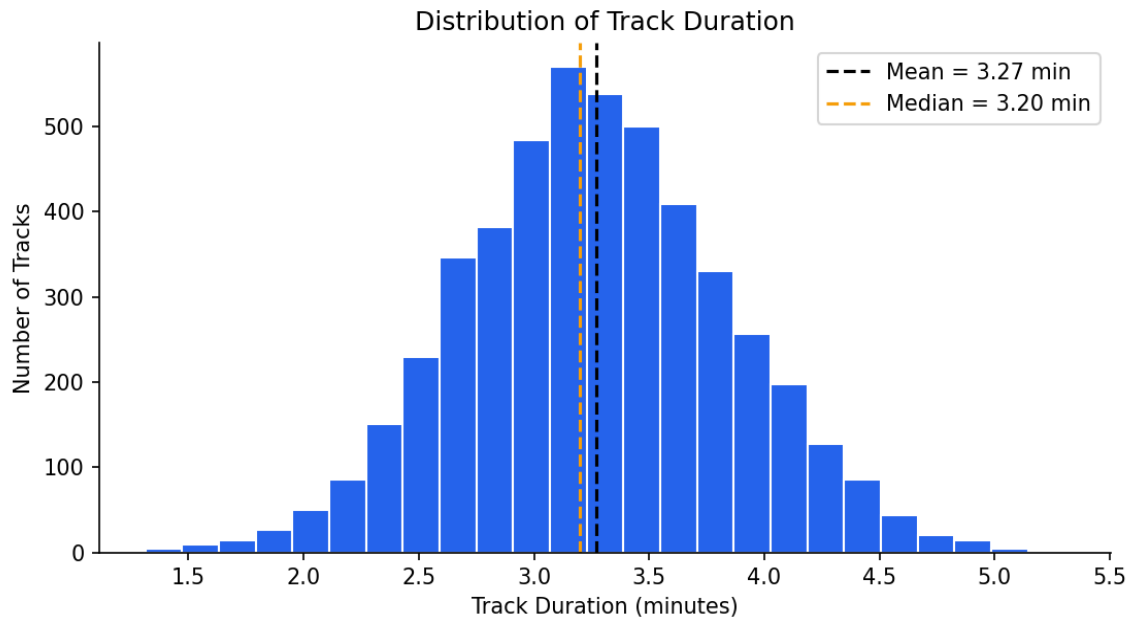


Figure 10. Distribution of track duration

INSIGHTS:

- Most tracks run between 2.5 and 4 minutes, consistent with streaming-era pack/duration norms.
- A smaller number of longer tracks (above 5 minutes) extend the tail of the distribution.
- Duration data can support playlist pacing and skip-rate planning alongside genre and popularity data.

KPI:

Median duration: ≈ 3.20 minutes

Mean duration: ≈ 3.27 minutes

15. Average Duration by Genre

The chart below compares average track duration across genres, giving a sense of which genres skew toward longer runtimes.

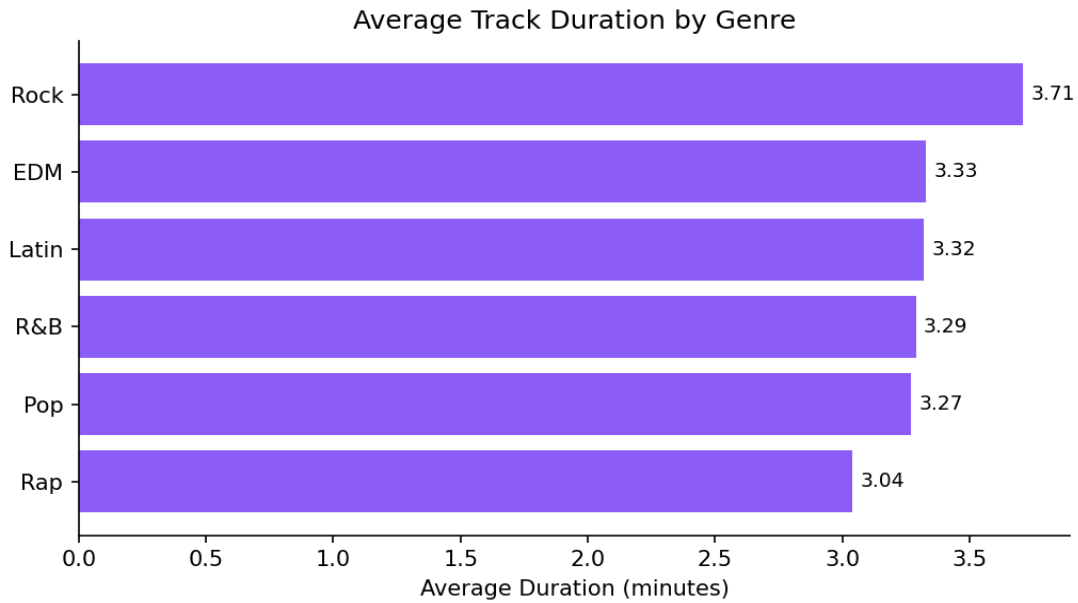


Figure 11. Average track duration by genre

INSIGHTS:

- Rock carries the longest average runtime (≈ 3.71 minutes), consistent with the genre's tradition of extended instrumental sections.
- Rap tracks are the shortest on average (≈ 3.04 minutes), a pattern widely attributed to streaming-era incentives that reward song length just long enough to count as a stream.
- Duration, genre and popularity are related: the shortest-running genre (rap) is also the highest-popularity genre.

KPI:

Longest genre (avg.): Rock (≈ 3.71 min)

Shortest genre (avg.): Rap (≈ 3.04 min)

16. Correlation of Audio Features with Popularity

To understand which audio characteristics move together with popularity, the chart below shows the correlation coefficient (r) between each audio feature and the track popularity score.

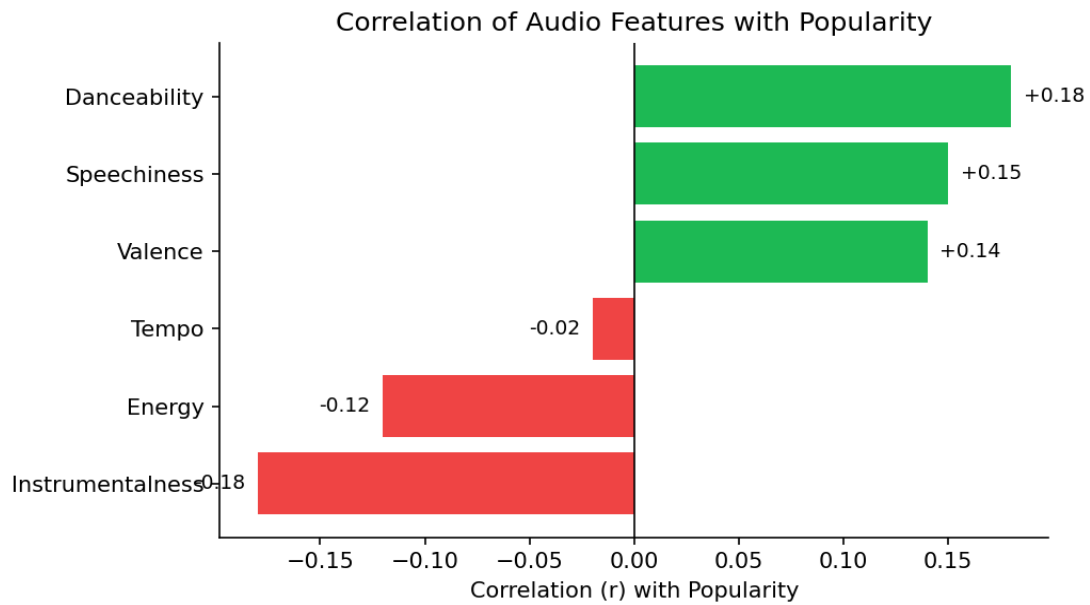


Figure 12. Correlation of audio features with popularity

INSIGHTS:

- Danceability ($r \approx 0.18$), speechiness ($r \approx 0.15$) and valence ($r \approx 0.14$) show the strongest positive associations with popularity.
- Instrumentalness ($r \approx -0.18$) and energy ($r \approx -0.12$) show the strongest negative associations.
- Tempo shows essentially no linear association with popularity ($r \approx -0.02$).
- These are descriptive associations only — none of the coefficients are strong enough to support a predictive claim on their own.

17. Cumulative Popularity Contribution by Genre

Summing each track's popularity score by genre gives an estimate of cumulative popularity contribution — useful for understanding which genres drive the most total listening interest, as distinct from average popularity per track.

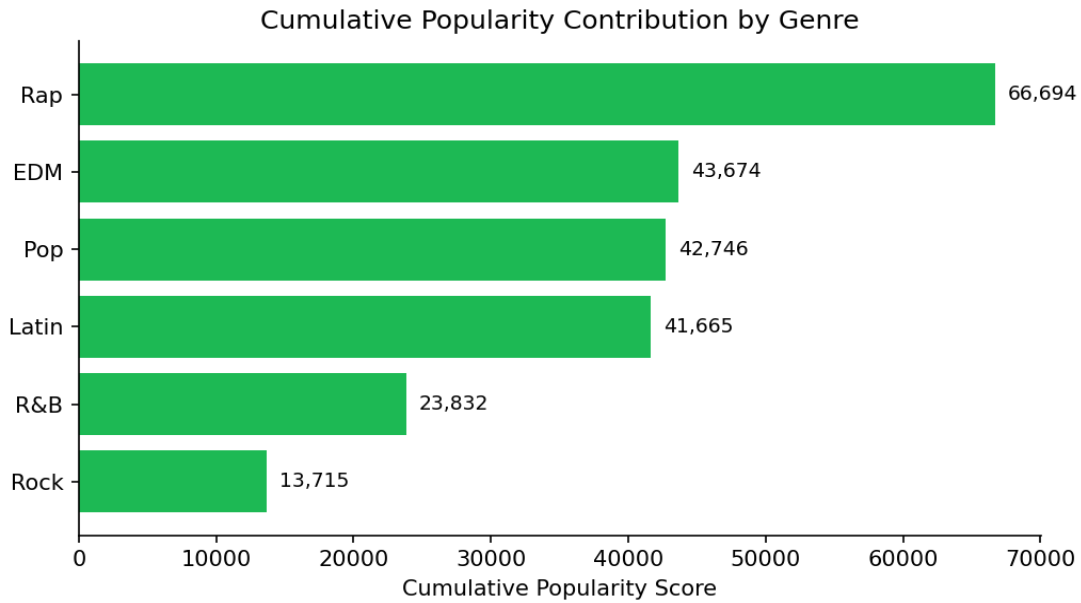


Figure 13. Cumulative popularity contribution by genre

INSIGHTS:

- Rap contributes the most cumulative popularity (66,694), consistent with its combination of high track volume and high average popularity.
- EDM contributes more cumulative popularity than its average-popularity ranking alone would suggest (43,674), reflecting its high track volume rather than high per-track quality.
- Rock contributes the least cumulative popularity (13,715), driven by both its small track count and its low average popularity.

KPI:

Total cumulative popularity: 232,326 across the catalog

Top contributor: Rap (66,694)

Lowest contributor: Rock (13,715)

18. Low-Popularity Tracks by Genre

Tracks scoring 0-20 (the Very Low tier) represent releases that gained little to no traction in the window. The chart below shows how many such tracks exist per genre.

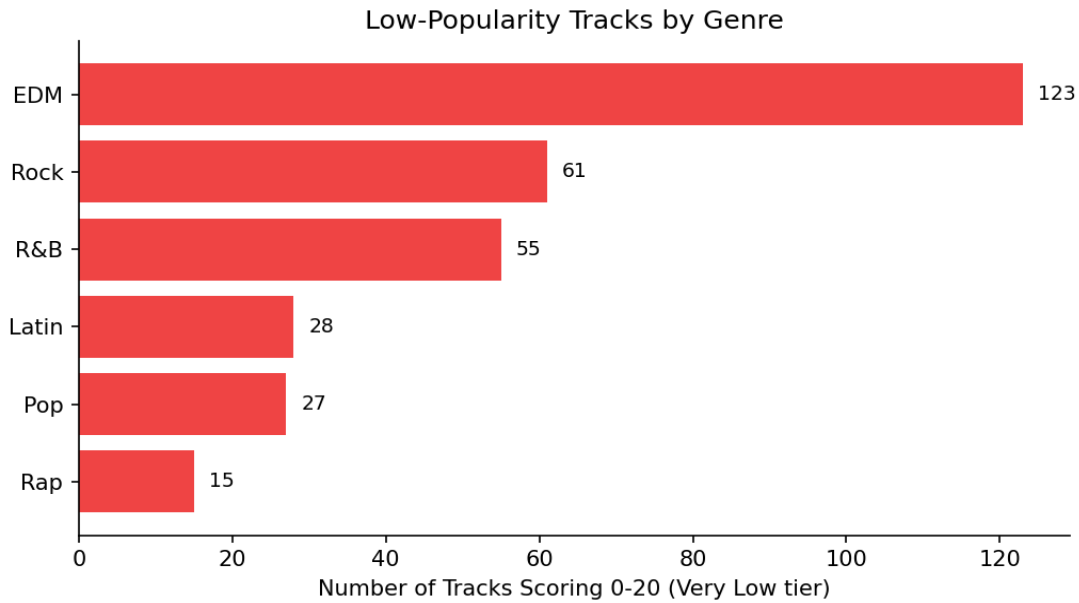


Figure 14. Low-popularity tracks by genre

INSIGHTS:

- 308 tracks — roughly 1 in 16 of the catalog window — scored in the Very Low (0-20) tier.
- EDM accounts for the largest share of low-popularity tracks (≈ 123), mirroring its large release volume.
- Rap shows the fewest low-popularity tracks (≈ 15), reflecting its tight concentration of tracks in the Moderate-to-High tiers.
- This list is a natural candidate for a review of release/promotion strategy, since these tracks gained the least traction despite being part of a major-label release cycle.

KPI:

Tracks in Very Low tier (0-20): 308 ($\approx 6.3\%$ of window)

Highest-risk genre: EDM (≈ 123 tracks)

19. Monthly Release Trend

Because each track carries an album release date, monthly release volume and average popularity can be tracked across the six-month window.

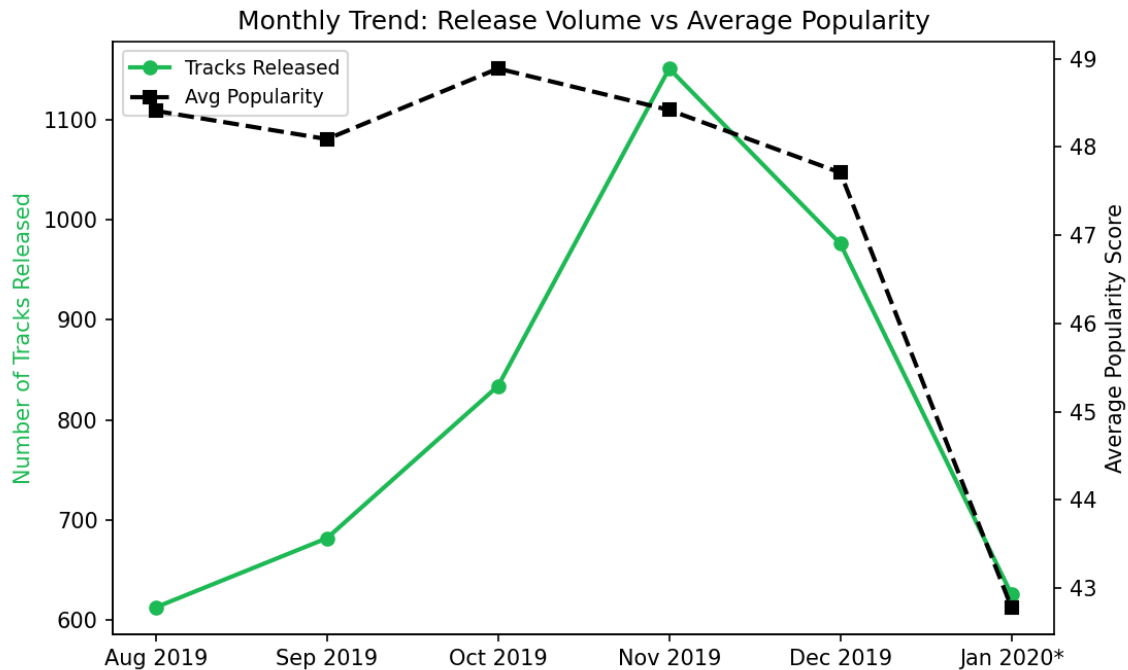


Figure 15. Monthly release volume vs average popularity

INSIGHTS:

- Release volume rose steadily from 613 tracks in August 2019 to a peak of 1,151 tracks in November 2019 — likely reflecting the industry's pre-holiday release push.
- Average popularity stayed remarkably stable, in the high 47-to-49 range, for the first five months, then dropped to 42.8 in January 2020, the lowest average of the window.
- January 2020 is a partial month (data collected through 29 January), and newer releases have had less time to accumulate streams — both likely contribute to the lower reading.

KPI:

Peak release month: November 2019 (1,151 tracks)

Lowest avg. popularity month: January 2020† (42.78)

20. Genre Share of Total Popularity

Viewing cumulative popularity as a share of the whole catalog window complements the per-track averages in Section 4 with a volume-weighted view.

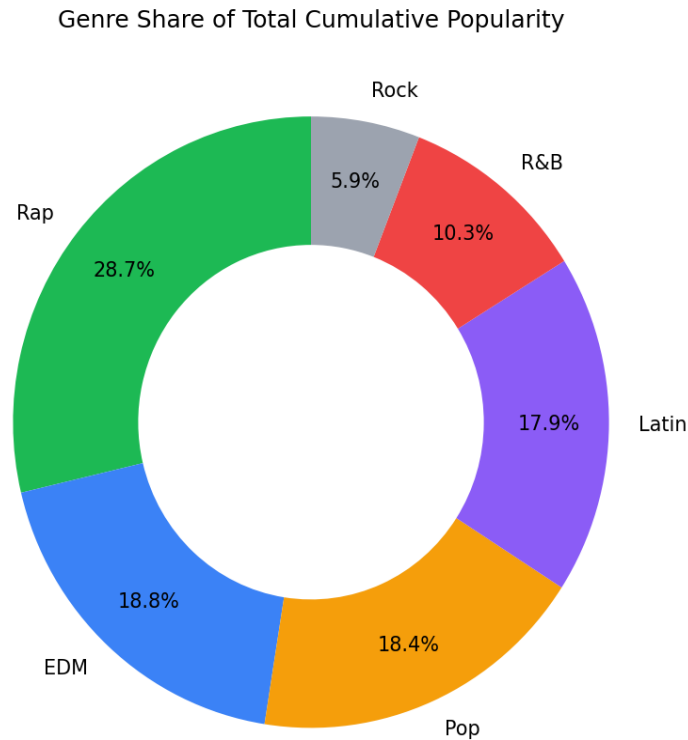


Figure 16. Genre share of total cumulative popularity

INSIGHTS:

- Rap accounts for the largest share of total catalog popularity ($\approx 28.7\%$), consistent with its high SKU-equivalent track count and above-average popularity per track.
- Rock contributes the smallest share ($\approx 5.9\%$), reflecting both its small track count and its lower average popularity.
- This volume-weighted view is useful for prioritising which genres matter most to overall catalog performance, as opposed to per-track averages alone.

Summary of Key Metrics

Metric	Value
Total tracks	4,882
Genres tracked	6
Overall avg. popularity	≈ 47.59
Overall median popularity	47.00
Overall avg. danceability	$\approx 67.4\%$
High/Very High-tier tracks (61-100)	1,158 ($\approx 23.7\%$)
Very Low-tier tracks (0-20)	308 ($\approx 6.3\%$)
Median track duration	≈ 3.20 minutes
Peak release month	November 2019 (1,151 tracks)
Top genre by cumulative popularity	Rap (66,694)

Next Steps to Move from Descriptive → Predictive

- Popularity forecasting by genre and release month once continuously-updated stream data is available.
- Breakout-hit prediction using audio-feature and release-timing patterns to flag tracks likely to reach the Very High tier early.
- Playlist/curation optimization linking genre and tier composition to actual listener engagement, once session-level data is added.
- Artist-cohort analysis once longer historical release windows become available.
- Cross-platform comparison (Spotify vs Apple Music, YouTube Music) once equivalent external data is accessible.

This report answers what the supplied catalog window looked like across August 2019-January 2020. The next analytical layer would combine this track-level data with continuous stream/engagement data to explain why popularity and genre patterns occur, and to forecast what is likely to happen next.

Glossary

Popularity score — Spotify's own 0-100 measure of how heavily a track is currently being streamed relative to all other tracks.

Danceability — A 0-1 Spotify audio feature describing how suitable a track is for dancing, based on tempo, rhythm stability, beat strength and overall regularity.

Energy — A 0-1 perceptual measure of intensity and activity in a track.

Valence — A 0-1 measure of the musical positiveness conveyed by a track.

Popularity tier — A grouping of tracks into five bands (Very Low, Low, Moderate, High, Very High) based on popularity score.

Breakout hit — A track reaching the High or Very High popularity tier (score ≥ 61).

Genre — A broad grouping of tracks (Pop, Rap, Rock, Latin, R&B, EDM) used for catalog organization and reporting.